MIL-S-87021B(NU)

23 January 1985

SUPERSEDING
MIL-S-87021A(SA)
12 January 1979

MILITARY SPECIFICATION

SHIRT, DISPOSABLE

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a disposable shirt.

1.2 Classification. The shirt covered by this specification shall be of one type and in the following sizes (see 6.2):

X-Small
Small
Medium
Large
X-Large
XX-Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

MILITARY

MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-C-87019	- Cloth. Nonwoven, Disposable Spunbonded Olefin (Coated and Uncoated)

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents drawings, and publications form a part of this specification to the extent specified herein

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W.. Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee Room 1106, 222 South Riverside Plaza, Chicago. Illinois 60606).

2.3 Order of precedence In the event of a conflict between this specification and the reference cited herein the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Cloths, nonwoven.

3.3.1.1 Cloth, nonwoven, disposable. The cloth for the back shall conform to the requirements for type I of MIL-C-87019.

3.3.1.2 Cloth, coated, nonwoven, disposable. The cloth for the front, sleeves, and collar shall conform to the requirements for type II class B of MIL-C-87019

3.3.2 Thread. The sewing thread shall be any thread suitable for the purpose intended.

3.3.3 Fastener, snap. The snap fastener shall consist of a stud (part No. 66-4100-00-00-6200) a socket (part No. 66-4200-00-00-6200) and prongs (part No. 66-4300-00-00-6200) or equal (see 6.4 and 6.5). The color shall be white.

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3.3.4 Label. Each shirt shall have a combination identification, size and cautionary statement label bearing the inscription indicated below. The size of the label shall be a minimum of 2 by 3 inches. The inscription shall be marked directly on the shirt by printing, stamping or stenciling or shall be printed on a paper label backed with a pressure sensitive adhesive for securing the label to the shirt. The cautionary statement shall be printed in larger letters than all other lettering on the label.

SHIRT, DISPOSABLE
CONTRACT NO.: DLA 100-00-0-000 (Example)
STOCK NO.: 8415-00-000-0000 (Example)
SIZE: X-Large (Example)
NAME OF CONTRACTOR:

CAUTION: THIS GARMENT SHALL NOT BE REUSED AND MUST BE DISCARDED WHEN OTTO FUEL STAINS ARE OBSERVED.

* 3.4 Design. The shirt shall be a below-the-hips length style, have full-length set in one or two piece sleeves, a sewn-on collar and a full front opening secured by 5 plastic snap fasteners. The collar, front and sleeves shall be of white polyethylene coated spun bonded olefin material and the back shall be of uncoated spun bonded olefin material. The coated side of the material is to be considered the face side.

3.4.1 Figure. The figure is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns to be used for cutting working patterns will be furnished by the Government (see 6.3). The patterns allow 3/8 to 1/2 inch for all seams. Neither the Government patterns nor the working patterns shall be altered in any way.

* 3.5.1 List of pattern parts. The component parts of the shirt shall be cut from the materials specified and in accordance with the pattern parts required for the manufacturing process as follows

Material	Nomenclature of pattern parts	Cut parts
Cloth, Nonwoven, Perforated, Type I	Back 1/	1
Cloth, Nonwoven, Coated, Nonperforated, Type II	Front	2
	Sleeve 2/	4
	Collar	1

1/ This part may be cut into two pieces with a 3/8 to 1/2 inch seam allowance added.
2/ This part may be cut as one piece.

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3.6 Construction.

3.6.1 Stitch, seam and stitching types. The stitch, seams and stitching types specified herein shall conform to FED-STD-751 Unless otherwise specified herein, the seam type shall be SSp or SSn-1 used with any 500 stitch type (except 501, 510, 511, or 513) or 1sc-2 with 301 or 401 stitch type.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be 6 and 8 respectively.

3.6.3 Thread breaks and ends of seams. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. A 1/4 to 1/2 inch of stitching chain shall remain at the end of the sleeve and side seams to prevent raveling Skipped stitches and thread breaks shall be repaired by stitching not less than 1/2 inch beyond each end of the defect using the 301 stitch.

3.7 Manufacturing operations requirements. The shirts shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished shirts are identical to those produced by following the sequence as shown.

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TABLE 1 - CONSTRUCTION OF SHIRT

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
1.	Cutting. The shirt shall be cut in strict accordance with the Government patterns furnished which show size, shape, directional lines for cutting, and notches for proper assembly of all parts. The face side of the coated basic material to be assembled shall be the coated side.					
2.	Replacement of defective components During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in Section 4 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Marking All parts shall be marked or bundled to insure proper assembly of the shirt. The use of metal devices, sew-on tickets or drill holes is not permitted.					
4.	Make shirt a. Stitch two-piece sleeves together face to face to form outseam.		1/		2/	3/
	b. Join fronts to back at shoulders face to face.					
	c. Insert sleeves and join sleeves to body of shirt.					

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TABLE 1 - CONSTRUCTION OF SHIRT

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl Bob/ Lpr
	d. Form facing and attach five (5) socket parts w/prong backing on the left front and five (5) stud parts w/prong backing on the right front where indicated by marks on pattern.				
	NOTE: As an alternate, the center of the top snap fastner shall be positioned 1/4 to 3/8 inch from the top of the neckline. The remaining snap fasteners shall be spaced equidistant, 5 1/2 ± 1/4 inches on center for all sizes. The distance from the folded edges of the facing shall be the same as that indicated by patterns.				
	e. with the collar positioned on the neckline, face side of collar to inside of jacket, stitch continuously across the entire neckline, include the top edges of each facing in the line of stitching.				
	f. Mark or adhere the label (see 3.3.4) to the inside of the shirt approximately 2 inches below the neckline and centered on the back.				
	1/ For seam and stitching types refer to paragraph 3.6.1.				
	2/ For stitches per inch, refer to 3.6.2				
	3/ For thread sizes refer to 3.3.2.				

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3.8 Sizes and measurements. The sizes and measurements of the finished shirts shall be as shown in Table II. All measurements and tolerances are expressed in inches.

Table II - Size and measurements

	A	B	C
Sizes	Chest	Inseam Sleeve Length	Center Back Length
X-Small	20-1/2	22-1/4	29-1/2
Small	21-1/2	22-1/4	30
Medium	23-1/2	22-5/8	30-1/2
Large	25-1/2	23-1/8	31
X-Large	27-1/2	23-1/2	31-1/2
XX-Large	28-1/2	23-7/8	32
Tolerance	± 1	+1/2, -3/8	+1/2, -3/8

NOTE: A, B, and C refer to Figure 1.

A. Chest - Measure across the chest at the base of the armhole from folded edge to folded edge.

B. Sleeve inseam - Measure along inseam from armhole to bottom of sleeve.

C. Length - Measure along center back from lower edge of neckline to bottom of shirt.

3.9 Workmanship. The finished shirt shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one finished shirt. The lot size shall be expressed in units of shirts.

4.4.2.1 Visual examination. The shirts shall be examined for defects classified in accordance with the list below.

Examine	Defect
Material Defects and Workmanship	Any evidence of delamination of the coating Any hole, needle chew, cut, tear, mend or burn.
Component and Assembly	Any component omitted, misplaced, damaged or otherwise defective. Any required operation omitted or improperly performed. Objectional odor
Seams and Stitching	Ends of a continuous line of stitching overlapped less than 1/2 inch. Thread breaks or two or more skipped or runoff stitches overstitched less than 1/2 inch beyond defective stitching area. Any open seam (i.e., one or more broken, skipped or runoff stitches or any raw edge not securely caught in a seam). Seam or stitch type not as specified Stitches per inch not as specified. Loose stitch tension resulting in a loose seam. Tight stitch tension (stitches break when normal strain is applied to the seam or stitching). Ends of a line of stitching when not continuous. Stitching chain runoff less than 1/4 inch or more than 1/2 inch.

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Examine (cont'd)	Defect
Label	Missing, incorrect, illegible or misplaced Cautionary statement lettering not larger than all other lettering. Not size specified.
Cleanness	Spot or stain Thread ends not trimmed

4.4.2.2 Size measurement defects.

a. Any measurement deviating from the dimensions and tolerances specified in Table II shall be scored as a size measurement defect.

b. Sleeve inseam lengths uneven by 1 inch or more shall be scored as a size measurement defect.

4.4.2.3 Inspection levels and acceptable quality levels. The inspection levels and acceptable quality levels (AQL's), expressed as defects per 100 units, for the visual and dimensional examinations shall be as follows

	Inspection Level	AQL
For examination in 4.4.2.1	II	2.5
For examination in 4.4.2.2	S-3	4.0

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and AQL shall be 2.5 defects per hundred units (DHU).

Examine	Defect
Marking (exterior)	Omitted, incorrect; illegible; of improper size. location sequence or method of application.
Materials	Any component missing. damaged, or not as specified.
Workmanship	Inadequate application of components. such as: Incomplete closure of container flaps, improper taping loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of shirts per shipping container is more or less than specified.

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4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size location, sequence or method of application.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C (See 6.2).

5.1.1 Level A. Each shirt shall be folded to measure approximately 14 by 11 inches.

5.1.2 Level C (Commercial Packaging). The shirts shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C (see 6.2).

5.2.1 Level A. Eighty (80) shirts of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF class weather-resistant, variety DW, grade V15c, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and forty in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

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5.2.2 Level B. Eighty (80) shirts of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and Reinforced conforming to type CF, class domestic, variety DW, grade 275, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and forty in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.3 Level C (Commercial Packing). Shirts, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes. Each shipping container, packed with mixed sizes only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained herein.

5.4 Palletization. When specified (see 6.2) (shirts), packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L or bonding means O. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

6. NOTES

6.1 Intended use. The shirt is intended primarily for use as part of an ensemble used to protect torpedo handlers from exposure to Otto fuels.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Sizes required (see 1.2).
c. First article requirements (see 3.2).
d. Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.4).

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6.3 Samples For access to samples, address the procuring activity issuing the invitation for bids.

6.4 Snap fasteners. The snap fastener components numbered 66-4100-00-00-6200, 66-4200-00-00-6200, and 66-4300-00-00-6200 are manufactured by Scovill Inc., Apparel Fastener Division, 500 Fifth Avenue, New York, N.Y. 10110. ATTN: Government Sales

6.5 "Or Equal" items Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

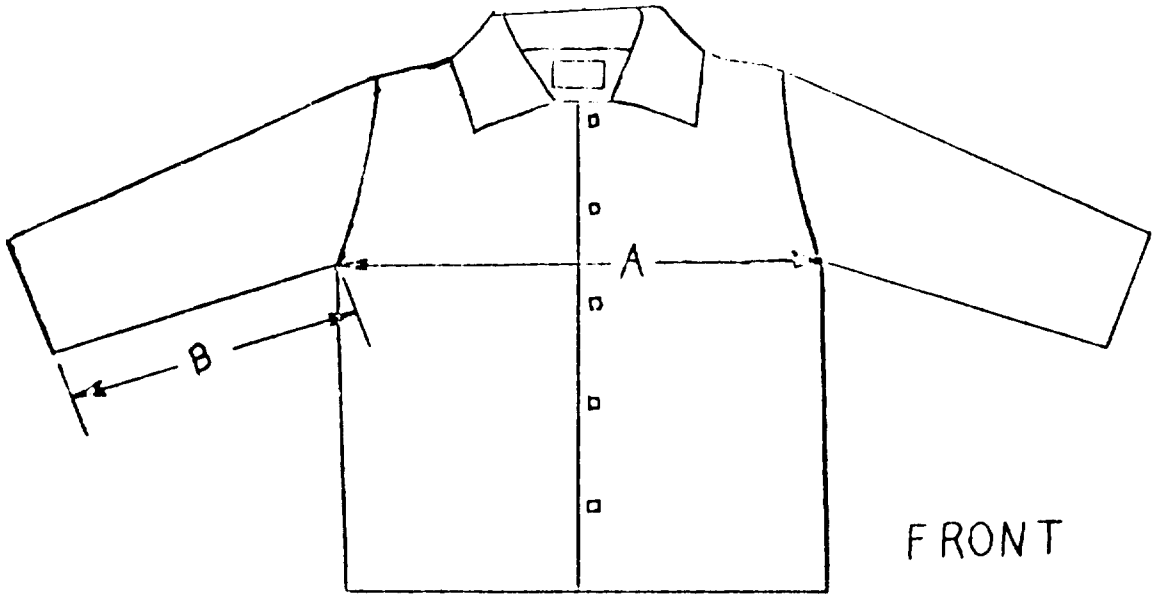
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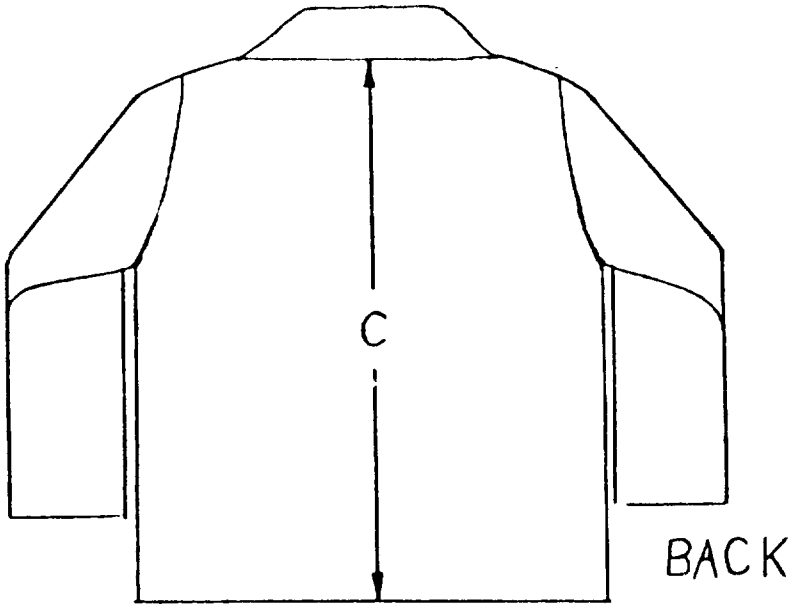


FIGURE 1 - SHIRT, DISPOSABLE

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-87021B(NU)	SHIRT, DISPOSABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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